

Real-Time Deep Learning Euro Coin Detection

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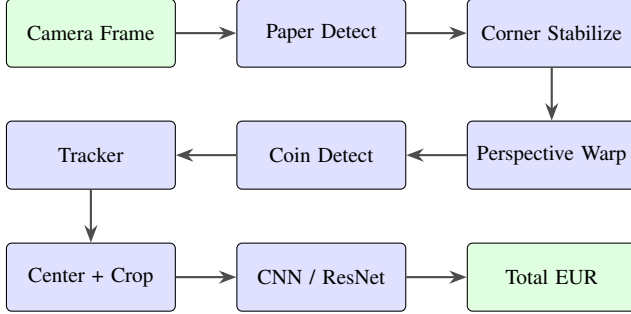


Fig. 1: End to end coin counting pipeline.

I. INTRODUCTION

This presentation extends the previous coin counting solution that relied on classical computer vision and an SVM classifier. While that system achieved reasonable accuracy, it had two key limitations: The Hough Circle Transform failed to separate touching coins reliably, and physical features (diameter and color) limited classification precision.

The present work solves both issues. Coin detection now employs **watershed segmentation**, which requires careful tuning, but delivers more robust separation of adjacent coins. Classification is managed by two deep learning architectures: a lightweight custom CNN and a ResNet, both trained on a custom dataset. Additionally, the source code was restructured around asynchronous queues for improved throughput.

The overall pipeline (Fig. 1) remains unchanged at a high level: frames are captured, the paper workspace is detected and rectified, coins are segmented and tracked, and each crop is classified to compute a running total.

II. DATASET ACQUISITION

A structured acquisition protocol was designed to capture coins under varying orientations, lighting, and positions. The warped workspace was divided into six labeled zones (Fig. 2), each assigned to a specific denomination. When the detector identified all coins in the scene, a screenshot and the corresponding labels (circle center and zone ID) were saved manually. Zone assignments were rotated after every capture to ensure spatial diversity in the resulting dataset.

Postprocessing consisted of cropping each coin using its detected center, producing 150×150 px labeled patches (Fig. 3). The dataset was further augmented with varied illumination conditions to improve generalization.

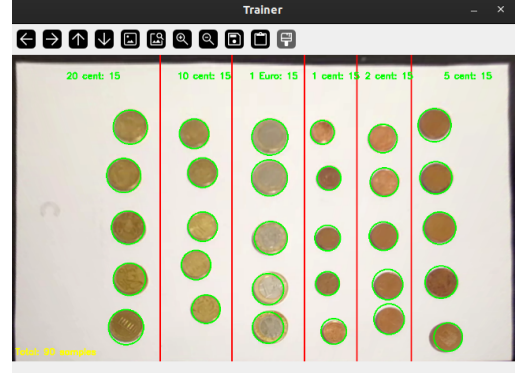


Fig. 2: Warped workspace with six labeled acquisition zones.

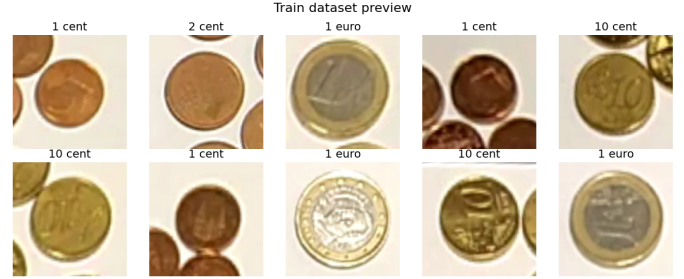


Fig. 3: Sample labeled coin crops from the collected dataset.

III. COIN DETECTION

The detection stage was redesigned around the watershed algorithm, replacing the Hough Circle Transform used previously. The process consists of three steps:

- 1) **Thresholding and morphology:** Binary segmentation of the coin regions followed by morphological opening to remove noise.
- 2) **Marker computation:** Distance transform based markers are generated to seed the watershed.
- 3) **Watershed and extraction:** The watershed transform separates overlapping coins. Detected contours are deduplicated and their centers extracted.

Although watershed requires more careful parameter tuning than Hough circles, it proves to be significantly more stable results when coins are touching.

IV. COIN CLASSIFICATION

Each detected coin is cropped to a 150×150 px patch and passed to a deep learning classifier. Two architectures were

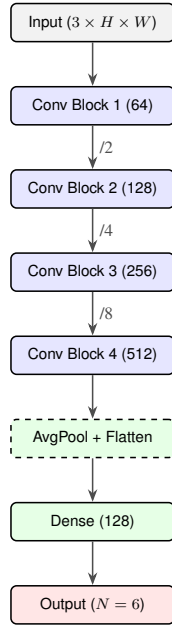


Fig. 4: Small CNN architecture.

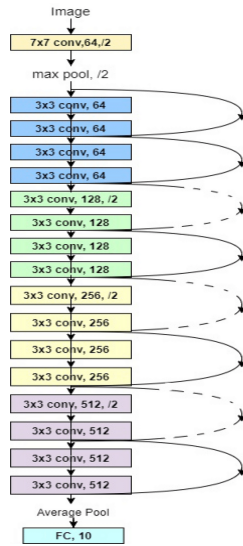


Fig. 5: ResNet architecture.

evaluated.

A. Architectures

Small CNN. A compact model (Fig. 4) with three convolutional blocks, each consisting of convolution, batch normalization, ReLU, and max pooling, followed by a fully connected head.

ResNet. A deeper residual network (Fig. 5) with skip connections, providing higher capacity at the cost of additional parameters.

B. Training and Evaluation

Both models were trained on the augmented dataset described in Section II. Figure 6 shows the training and val-

idation curves; Figure 8 presents the confusion matrices on the held out test set. The ResNet model achieves near perfect classification across all denominations, while the smaller CNN offers a favorable accuracy to latency trade off for real-time deployment, but because one single coin can ruin the count, it is critical to keep with ResNet.

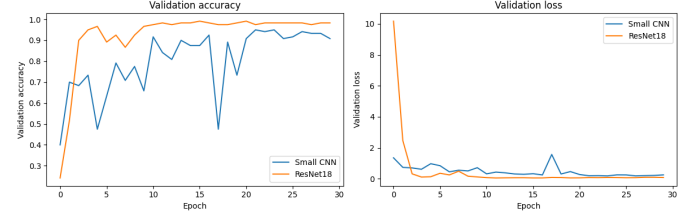


Fig. 6: Training and validation accuracy/loss for both models.

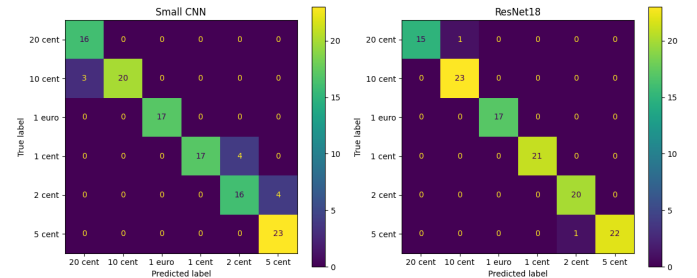


Fig. 7: Confusion matrices: Small CNN (left) vs. ResNet18 (right).

Finally an inference test was made to have more information of possible bottlenecks or evaluate performance.

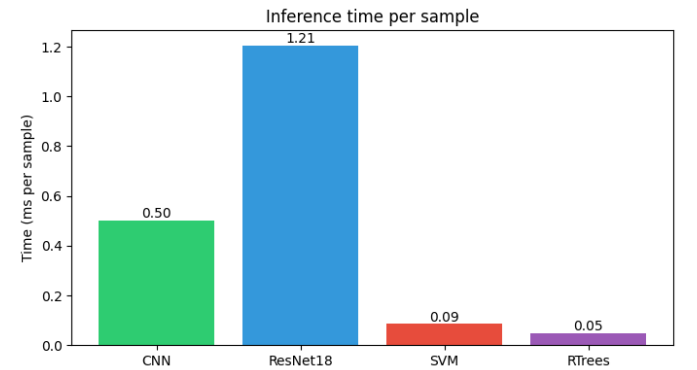


Fig. 8: Inference times for each classifier

V. RESULTS

The complete pipeline runs in real time on a standard desktop. A video demonstration of the system detecting, tracking, and counting euro coins is available at:

Video Demonstration
https://youtu.be/ZT53_QFDRCY